By the end of this session, students will be able to…

- Identify the clinical findings/presentations when a patient has a urinary tract infection
- Develop a differential diagnosis in regard to infections of the urinary tract
- Develop a plan for ordering tests using evidence-based guidance and understand how to interpret those laboratory tests
- Develop a treatment plan for patients with a specific urinary tract infections
- Identify patient populations that are at specific risk of adverse events when an infection of the urinary system occurs

Source: Course Syllabus

Asymptomatic Bacteriuria

- Absence of symptoms attributable to the bacteria in the urinary tract and does not usually require treatment.
- Incidentally found on screening urine culture.

Diagnosis:

- Women:
  - Clean-catch voided specimens on two separate occasions must contain $>10^5$ CFU/mL of the same bacterial strain, **or**
  - One catheterized specimen must contain $>10^2$ CFU/mL of bacteria.
- Men:
  - A single clean-catch specimen with $>10^5$CFU/mL of bacteria **or**
  - One catheterized specimen with $>10^2$ CFU/mL of bacteria.

Urinary Tract Infections, Cystitis, Prostatitis, and Pyelonephritis, *Harrison's Principles of Internal Medicine, 22e*

Asymptomatic Bacteriuria

- **Screen and treat pregnant women** (12-16 weeks gestation)
  - 7-day course of amoxicillin
  - Penicillin allergy: 7-day course of Nitrofurantoin or a cephalosporin

Case #1

A 29-year-old woman complains of a 2-day history of dysuria, urgency, and urinary frequency. She denies the use of medications and has no significant past medical history. On examination, her blood pressure (BP) is 100/70 mm Hg, heart rate (HR) is 90 beats per minute, and temperature is 98°F (36.6°C). The heart and lung examinations are normal. She does not have back tenderness. The abdomen is nontender and without masses. The pelvic examination reveals normal female genitalia. There is no adnexal tenderness or masses.

What is the most likely diagnosis?

What is the next step in the diagnosis?

Uncomplicated Bacterial Cystitis

- Most common in women

- Diagnosis
  - Dysuria
  - Frequency, urgency, or both
  - Cloudy, smelly urine
  - Urine dipstick analysis positive for nitrites or leukocyte esterase.
  - Positive urine culture (>10^4 organisms).
  - **No fever or flank pain**

Uncomplicated Bacterial Cystitis [Young Women]

Factors that promote cystitis	Factors that prevent cystitis	Factors with no evidence of effect on cystitis
• Sexual activity and practices • Genetic predisposition • Delayed postcoital voiding	• Cranberry juice • Prophylactic antibiotics	• Precoital voiding • Underwear fabric • Douching • Hot tub use

Basic guidelines – Treatment of UTI

- Treatment is directly related to level of infection
- In most females, treatment may be started without testing
- Any other case, urine culture must be done. Use empirical antibiotics while waiting
- In male and children, follow-up investigation must be done after first UTI
- Differential diagnosis always includes STI

Acute Cystitis - Treatment

Uncomplicated

- **Trimethoprim-sulfamethoxazole** x **3** days

Complicated

- **TMP/SMX** Long course (**≥ 10-14** days)

Sulfa allergy

- **Nitrofurantoin** x **5-7** days or
- **Fluoroquinolone** x **3** days
  - Levofloxacin
  - Ofloxacin
  - Ciprofloxacin

Pregnant women

- **Amoxicillin**, **cephalosporin**, **nitrofurantoin**, or another pregnancy-safe antibiotic x **7** days

Urethritis

Diagnosis

- Pain or irritation on urination
- No frequency or urgency
- No or low level suprapubic tenderness
- + Discharge from the urethra (predominantly males)
- Vaginal discharge possible
- Low bacterial count or “sterile urine” on culture

E. coli UTI favored if:

- Gross hematuria
- Abrupt onset
- Duration < 3 days
- Hx of previous UTI

Rule out STI

Acute Pyelonephritis

- Clinical Findings:
  - Mild: low-grade fever without or with lower-back or costovertebral angle (CVA) tenderness
  - Severe high fever, rigors, nausea, vomiting, flank and/or loin pain
  - **Fever** is the main feature distinguishing cystitis and pyelonephritis

- Diabetic patients: obstructive uropathy associated with acute papillary necrosis, emphysematous pyelonephritis

Acute Pyelonephritis

- May be managed as outpatient
- Hospitalize if:
  - Intractable vomiting
  - Evidence of shock
  - Severe dehydration
- Initiate treatment in the hospital for pregnant females
- Treatment should last at least 10 days
- Follow-up 5 days after treatment completion

Acute Pyelonephritis - Treatment

- Obtain urine culture, then start empiric treatment

- Bactericidal, broad spectrum
  - Fluoroquinolones
  - Aminoglycosides
  - Aminopenicillins (amoxicillin with clavulanic acid, ticarcillin, or piperacillin)
  - Cephalosporins
  - Imipenem

- Treat for 10 days of coverage
  - May need to start IV antibiotics if PO not tolerated

Acute Pyelonephritis - Treatment

- Special Considerations in Pregnancy
  - Initiate treatment with IV or IM in hospital
  - When afebrile, patient may take oral meds to complete 10 days treatment
  - Ensure eradication of organism with repeat urine culture 5 days post treatment

Pyelonephritis

- Indications for imaging
  - Uncertain diagnosis
  - Severe illness or immunocompromised status
  - Worsening renal function
  - Suspected kidney stone or urinary tract obstruction
  - Relapsing or recurrent pyelonephritis
  - Male gender
  - Failure to improve after 72 hours of antibiotics

Consensus Classification of Prostatitis

Syndrome	Description	WBC count per HPF (urine)	Bacterial Cultures	Special Tests
Acute Bacterial Prostatitis	Acute infection of the prostate gland	>10	+ on voided specimen	
Chronic Bacterial Prostatitis	Chronic infection of the prostate gland	>10	(-) on voided specimen	Expressed prostatic secretions will be +
Chronic Pelvic Pain Syndrome	Chronic pelvic pain in the absence of bacteria localized to the prostate	>10 (In non-inflammatory causes there will be <10)	(-) bacteria on any specimen	

Acute Bacterial Prostatitis

Symptoms:

- Dysuria, frequency, and urgency
- Low back, perineal, penile, or rectal pain
- Fever and chills

Physical Exam:

- Tense/boggy prostate
- Purulent discharge
- Urine culture positive
  - Most commonly gram-negative organism (*E. coli or Klebsiella)*

Nursing Homes/ Long Term Care Settings

Prevalence of UTI reported 15-50% in LTC settings

- Likely to be higher than current estimates

Special Issues

- Multidrug resistant bacteria and overgrowth of organisms such as *C difficile*
- *Pseudomonas, enterococci, staphylococci*, and *candida* species are more common than in the outside population
- E. coli is seen in 50% (however tend to be resistant strains)

Use urine culture and sensitivity

Treat using narrow spectrum antibiotics

Clinical decision making for:

When to refer to a specialist:

- Anatomic abnormalities leading to repeated urinary infections
- Infections associated with nephrolithiasis
- Persistent interstitial cystitis/painful bladder syndrome

When to admit the non-pregnant patient:

- Severe pain requiring parenteral medication or impairing ambulation or urination
- Dysuria associated with urinary retention or obstruction
- Pyelonephritis with ureteral obstruction

Summary - Key Questions to Consider

- Which patients with urinary tract infections need hospital admission?
- When should catheter-associated urinary tract infections be treated?
- When is renal imaging indicated?
- When should urology consultation be obtained?
- What are the appropriate durations of therapy for uncomplicated cystitis, catheter-associated urinary tract infections, and pyelonephritis?
- For which patients are follow-up urine cultures indicated after discharge?